MODULE 6

Process Simulation (Calculator)

Worksheet · worked example · sensitivities

Read this first – simplified estimate, not plant data
This module is a **pencil-and-calculator simulator**. It uses physics (stoichiometry, Faraday's law, ideal compression) plus a few published reference values (IRENA 2020). Items marked **[assumption]** are teaching inputs chosen for illustration – they are **not** vendor guarantees or real project costs. Replace them with OEM data and actual tariffs for any real study.

6.1 Inputs you can change

Input	Symbol	Example value	Basis
Electrolyser capacity	P (MW)	10 MW	[assumption] – your choice
Specific energy consumption	SEC (kWh/kg)	51.2 kWh/kg (= 65 % LHV)	IRENA 2020 reference case [R1]
Operating hours (renewable availability)	h (h/yr)	4,000 h/yr	[assumption] e.g. solar + wind hybrid
Electricity price	p (USD/MWh)	30 USD/MWh	[assumption]; IRENA uses 20 USD/MWh in sensitivities [R1]
System capex	C (USD/kW)	750 USD/kW	Mid-range of IRENA 2020 AEL 500–1,000 [R1]
Discount rate / life	r, n	8 % / 20 yr	8 % per IRENA [R1]; life [assumption]
Fixed O&M	–	2 % of capex per yr	[assumption]
Water use / cost	–	20 kg/kg; 2 USD/m³	IRENA 18–24 kg/kg [R1]; price [assumption]
Delivery pressure	–	30 → 350 bar	[assumption] for compression example

6.2 Equations

H_2 per hour (kg/h) = P × 1,000 ÷ SEC

H_2 per day = kg/h × 24 (at full load) · per year = kg/h × h

Electricity (MWh/yr) = P × h

O_2 (kg) = 8 × H_2 (kg) · Stoichiometric water = 9 × H_2
from 2H2O → 2H2 + O2 (36 : 4 : 32)

Efficiency (LHV) = 33.3 ÷ SEC · Efficiency (HHV) = 39.4 ÷ SEC

$CRF = r(1+r)^n \div [(1+r)^n - 1]$
capital recovery factor

LCOH = (C·P·1000·CRF + O&M) ÷ annual kg + SEC × p ÷ 1,000 + water
levelised cost of hydrogen, USD/kg

Ideal isothermal compression work = $(RT/M)\cdot\ln(P_2/P_1)$
divide by a realistic compressor efficiency

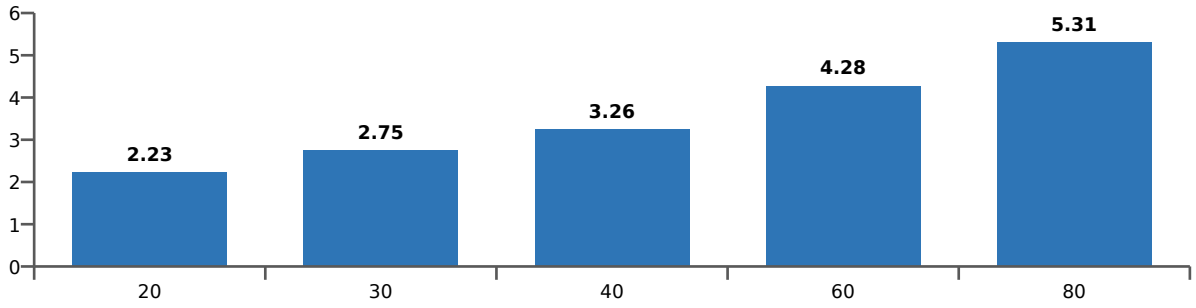
6.3 Worked example (10 MW, 4,000 h/yr)

Output	Calculation	Result
H_2 per hour	10 × 1,000 ÷ 51.2	195 kg/h
H_2 per day (full load)	195 × 24	4.69 t/day
H_2 per year	195 × 4,000	781 t/yr
Electricity	10 MW × 4,000 h	40,000 MWh/yr
Oxygen by-product	8 × H_2	6250 t/yr
Water (incl. treatment)	20 × H_2	15625 m³/yr
Efficiency	33.3 ÷ 51.2 (LHV) · 39.4 ÷ 51.2 (HHV)	65 % LHV · 77 % HHV
Capex share of LCOH	750×10,000×CRF ÷ annual kg	0.98 USD/kg
Fixed O&M	2 % × capex ÷ annual kg	0.19 USD/kg
Electricity	51.2 × 30 ÷ 1,000	1.54 USD/kg
Water	0.020 m³ × 2 USD	0.04 USD/kg
LCOH (simplified)	sum	**2.75 USD/kg**
Compression 30→350 bar (ideal)	(RT/M)·ln(350/30)	0.84 kWh/kg (≈1.7 kWh/kg at 50 % efficiency [assumption])

6.4 Sensitivity: LCOH (USD/kg) vs. electricity price and operating hours

Hours \ Price →	20 USD/MWh	30 USD/MWh	40 USD/MWh	60 USD/MWh	80 USD/MWh
2000 h/yr	3.40	3.92	4.43	5.45	6.48
3000 h/yr	2.62	3.14	3.65	4.67	5.70
4000 h/yr	2.23	2.75	3.26	4.28	5.31
6000 h/yr	1.84	2.36	2.87	3.89	4.92
8000 h/yr	1.65	2.16	2.67	3.70	4.72

Simplified LCOH at 4,000 h/yr vs electricity price (USD/MWh)



Lesson: electricity price drives cost once hours are high; at low hours the capex share rises steeply. This matches IRENA's finding that electricity is the largest cost component and capex the second [R1].

6.5 Worksheet – your own case

Input / output	Your value	Notes
Capacity P (MW)		
SEC (kWh/kg)		
Hours (h/yr)		
Electricity price (USD/MWh)		
Capex (USD/kW)		

Input / output	Your value	Notes
H2 per hour (kg/h)		
H2 per year (t)		
Electricity (MWh/yr)		
O2 (t/yr)		
Water (m^3/yr)		
Efficiency LHV / HHV		
LCOH (USD/kg)		

Simplified vs real performance

- Real SEC rises with stack **degradation** and at **part load** (balance-of-plant loads stay fixed).
- Stack replacement (IRENA reference: 80,000 h stack life [R1]) adds a mid-life cost not modelled here.
- Purity (de-oxo + drying) and compression add electricity; storage and transport add cost (Module 8).
- Taxes, land, grid charges, financing structure and subsidies (e.g. India SIGHT incentives) are excluded.

Quiz & revision – Module 6

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. 10 MW at 50 kWh/kg makes how much H2 per hour?
a) 20 kg **b)** 200 kg **c)** 2,000 kg **d)** 500 kg

Q2. Oxygen produced per kg H2:
a) 1 kg **b)** 2 kg **c)** 8 kg **d)** 16 kg

Q3. SEC of 51.2 kWh/kg equals what LHV efficiency?
a) ~45 % **b)** ~65 % **c)** ~77 % **d)** ~90 %

Q4. At low operating hours, which LCOH component grows most?
a) Electricity **b)** Capex share **c)** Water **d)** Oxygen credit

Q5. Largest cost component of green H2 (IRENA):
a) Water **b)** Electricity **c)** Land **d)** Compression

Answer key

Q1 – b: 10,000 ÷ 50; **Q2 – c**: 32 ÷ 4; **Q3 – b**: 33.3 ÷ 51.2; **Q4 – b**: fixed cost spread over fewer kg; **Q5 – b**: [R1]